How Many Generalised Killing Spinors Force a Hypersurface?

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Hypersurface Data and Gauss–Codazzi

Let $(M^n, g) \hookrightarrow (Z^{n+1}, \bar{g})$ be an oriented Riemannian hypersurface with unit normal ν:

$$TZ|_M = TM \oplus \mathbb{R}\nu, \qquad g = \bar{g}|_{TM}, \qquad W(X) := -\bar{\nabla}_X \nu.$$

The Gauss and Codazzi equations relate (g, W) to the ambient curvature:

$$R - W \wedge W = (\bar{R})^{\top}, \qquad (\nabla_X W)(Y) - (\nabla_Y W)(X) = -(\bar{R}(X,Y)\nu)^{\top}.$$

In Euclidean space $\bar{R} = 0$, so the compatibility conditions become

$$\mathcal{G}_W := R - W \wedge W = 0, \qquad \mathcal{C}_W(X,Y) := (\nabla_X W)(Y) - (\nabla_Y W)(X) = 0.$$

Fundamental Theorem of Hypersurfaces

For symmetric W: $\mathcal{G}_W = \mathcal{C}_W = 0$ if and only if (g, W) is locally realised by an isometric immersion $M \to \mathbb{R}^{n+1}$ with shape operator W.

The Spinor Bundle

On an oriented Riemannian spin manifold (M^n, g), there is a complex vector bundle

$$\Sigma M \longrightarrow M, \qquad \Sigma_p M \simeq \Delta_n, \qquad \operatorname{rank}_{\mathbb{C}} \Sigma M = \dim_{\mathbb{C}} \Delta_n = 2^{\lfloor n/2 \rfloor},$$

whose fibres are copies of the spin representation Δ_n; sections $\psi \in \Gamma(\Sigma M)$ are spinor fields. It carries two structures used here:

- A connection $\nabla^{\Sigma M}$, induced by the Levi-Civita connection;
- Clifford multiplication by tangent vectors, satisfying

$$X \cdot Y \cdot \psi + Y \cdot X \cdot \psi = -2g(X, Y)\psi.$$

In particular $X \cdot X \cdot \psi = -|X|^2 \psi$, so Clifford multiplication by a non-zero vector is invertible.

Parallel Spinors

A parallel spinor satisfies

$$\nabla^{\Sigma M}\psi = 0.$$

Ricci-flatness and special holonomy

One non-zero parallel spinor forces $\mathrm{Ric} = 0$, and such spinors occur naturally in Calabi–Yau, hyperkähler, G_2, and $\mathrm{Spin}(7)$ geometry.

- On $\mathbb{R}^m$, the spinor bundle is globally trivial,

$$\Sigma\mathbb{R}^m \cong \mathbb{R}^m \times \Delta_m,$$

  and the constant functions $\psi(x) \equiv \psi_0 \in \Delta_m$ give all parallel spinors.

Flat space attains the maximum possible count:

$$\dim_{\mathbb{C}} \mathcal{P}(\mathbb{R}^m) = 2^{\lfloor m/2 \rfloor}.$$

Real Killing Spinors

A real Killing spinor satisfies

$$\nabla_X \psi = \lambda X \cdot \psi, \qquad \lambda \in \mathbb{R}.$$

- Integrability of this overdetermined first-order system gives

$$\mathrm{Ric} = 4\lambda^2 (n-1) g.$$

- Thus a real Killing spinor forces g to be Einstein:

$$\mathrm{Ric} = cg.$$

Setting $\lambda = 0$ recovers the parallel case:

$$\lambda = 0 \quad \Longrightarrow \quad c = 0.$$

The Cone Correspondence

There is a way to turn Killing spinors back into parallel spinors. Form the cone

$$C(M) = \big((0,\infty) \times M,\ dr^2 + r^2 g\big).$$

Bär's theorem [Bär93]: after normalising $|\lambda| = 1/2$,

real Killing spinor on (M, g) $\quad\Longleftrightarrow\quad$ parallel spinor on $C(M)$.

Some familiar geometries are also naturally described by their cones:

Geometry of M		Geometry of $C(M)$
round sphere S^n	$\longleftrightarrow$	flat $\mathbb{R}^{n+1} \setminus \{0\}$
Sasakian	$\longleftrightarrow$	Kähler
Sasaki–Einstein	$\longleftrightarrow$	Ricci-flat Kähler (Calabi–Yau)
3-Sasakian	$\longleftrightarrow$	hyperkähler

The Spinorial Gauss Formula

Why does the cone produce the Killing equation? It is a hypersurface phenomenon. For a hypersurface $M^n \subset Z^{n+1}$ with shape operator W, an ambient parallel spinor restricts, under the hypersurface spinor identification, by the spinorial Gauss formula:

$$0 = \nabla^{\Sigma Z}_X \Psi = \nabla^{\Sigma M}_X \psi + \frac{1}{2} W(X) \cdot \psi.$$

Hence

$$\boxed{\nabla^{\Sigma M}_X \psi = -\frac{1}{2} W(X) \cdot \psi.}$$

Cone check: the slices $\{r\} \times M$ have $W = -\frac{1}{r}\,\mathrm{id}$, so $\nabla_X \psi = \frac{1}{2r} X \cdot \psi$: a real Killing spinor, with $\lambda = \frac{1}{2}$ on the unit slice, i.e., Bär's normalisation.

$$\begin{aligned} W = 0 \quad &\Rightarrow \quad \text{parallel spinor}, \\ W = c\,\mathrm{id} \quad &\Rightarrow \quad \text{real Killing spinor}, \\ W \text{ symmetric} \quad &\Rightarrow \quad \textbf{generalised Killing spinor (GKS)}. \end{aligned}$$

The Counting Question

For fixed symmetric W, define

$$\mathcal{K}_W(M) = \left\{ \psi \in \Gamma(\Sigma M) : \nabla_X \psi = -\frac{1}{2} W(X) \cdot \psi \quad \forall X \in TM \right\}, \qquad k := \dim_{\mathbb{C}} \mathcal{K}_W(M).$$

These are the parallel sections of $\nabla^W_X = \nabla^{\Sigma M}_X + \frac{1}{2} W(X)\cdot$, hence evaluation at one point is injective:

$$k \leq \operatorname{rank}_{\mathbb{C}} \Sigma M = 2^{\lfloor n/2 \rfloor}.$$

For a Euclidean hypersurface with its induced spin structure, restriction of the relevant ambient (half-)spinors gives the maximal value $k = \operatorname{rank}_{\mathbb{C}} \Sigma M$.

Question

How many common generalised Killing spinors, satisfying the same equation for W, are enough to force the Euclidean Gauss–Codazzi equations $\mathcal{G}_W = \mathcal{C}_W = 0$?

For surfaces, one GKS suffices: Friedrich showed that $k = 1$ already reconstructs a local isometric immersion into $\mathbb{R}^3$ [Fri98]. For $n = 3$, see also [LR08].

Heisenberg Says No

On the Heisenberg group H_3, take the left-invariant metric with orthonormal frame

$$[e_1, e_2] = a\, e_3, \qquad [e_1, e_3] = [e_2, e_3] = 0, \qquad a > 0.$$

Every invariant spinor satisfies the GKS equation with the same

$$W = \tfrac{a}{2}\,\mathrm{diag}(1, 1, -1), \qquad \text{so} \qquad k = 2 = \mathrm{rank}_{\mathbb{C}}\, \Sigma H_3 \quad [\text{Art25}].$$

But the sectional curvatures disagree with Gauss:

	K_{12}	K_{13}	K_{23}
Actual:	$-3a^2/4$	$a^2/4$	$a^2/4$
Gauss requires:	$a^2/4$	$-a^2/4$	$-a^2/4$

Even the full rank of ΣM can fail Gauss–Codazzi. Multiplicity alone is not enough in dimension three.

Main Theorem

Theorem

Let $n \geq 4$, let (M^n, g) be connected and spin, and let W be smooth and symmetric. If

$$k = \dim_{\mathbb{C}} \mathcal{K}_W(M) > \frac{1}{2} \operatorname{rank}_{\mathbb{C}} \Sigma M \qquad \left(\text{equivalently } k \geq 2^{\lfloor n/2 \rfloor - 1} + 1\right),$$

then $\mathcal{G}_W = \mathcal{C}_W = 0$, so (g, W) is locally realised by an isometric immersion into $\mathbb{R}^{n+1}$.

The threshold is sharp: half the spinor rank can occur without Gauss–Codazzi.

Why does the critical fraction 1/2 appear?

Wang's Classification

Parallel spinors are fixed by parallel transport around loops; these maps form the holonomy group. Wang's classification for irreducible non-flat Ricci-flat cases [Wan89]:

Holonomy	Dimension	Parallel Spinors	Efficiency
$SU(r)$	$2r$	2	2^{1-r}
$Sp(r)$	$4r$	$r+1$	$(r+1)/2^{2r}$
G_2	7	1	$1/8$
$\mathrm{Spin}(7)$	8	1	$1/16$

Compare each factor with the number of parallel spinors in Euclidean space of the same dimension:

$$\eta(Y) = \frac{\dim_{\mathbb{C}} \mathcal{P}(Y)}{\dim_{\mathbb{C}} \mathcal{P}(\mathbb{R}^{\dim Y})}, \qquad \max_{\text{non-flat}} \eta = \frac{1}{2} \quad \text{at} \quad \mathrm{Hol} = Sp(1) = SU(2).$$

Following the normalisation of Alekseevsky–Cortés [AC09], we call η the efficiency.

The Non-flat Ceiling and an Extremal Example

A flat ambient produces Euclidean hypersurface data. The counterexample therefore needs a non-flat, Ricci-flat, ambient with as many parallel spinors as possible.

Efficiency is multiplicative, and flat factors have efficiency one. Hence:

$$\boxed{Z^{n+1} \text{ non-flat Ricci-flat} \quad \Longrightarrow \quad \dim_{\mathbb{C}} \mathcal{P}_\sigma(Z) \;\leq\; 2^{\lfloor n/2 \rfloor - 1}}$$

($\mathcal{P}_\sigma$: the sign-relevant parallel spinors.)

- Use four dimensions for non-flat geometry: X^4 non-flat hyperkähler (e.g. a K3 surface with Yau's Ricci-flat metric),

$$\mathrm{Hol}(X) = Sp(1), \qquad \dim_{\mathbb{C}} \mathcal{P}(X) = 2.$$

- Keep all remaining directions flat: $Z = X^4 \times \mathbb{R}^{n-3}$.
- Take

$$M^n = X^4 \times S^{n-4}(r) \subset Z \qquad (n \geq 5).$$

Restricting the ambient parallel spinors gives $2^{\lfloor n/2 \rfloor - 1}$ common GKS on M.

A Half-Rank Counterexample

The failure of Gauss is visible in the normal: at $(x, y) \in X \times S^{n-4}(r)$,

$$\nu = \left(0, \tfrac{y}{r}\right) \in T_x X \oplus \mathbb{R}^{n-3},$$

with no X-component and constant in the X-directions. Hence

$$W|_{TX} = 0, \qquad W|_{TS^{n-4}(r)} = -\tfrac{1}{r}\,\mathrm{id},$$

while $R|_{TX} = R^X \neq 0$. Euclidean Gauss would require

$$R|_{TX} = (W \wedge W)|_{TX} = 0 :$$

the entire defect comes from the curvature of the X-factor.

For $n = 4$, we can take $M^4 = X^4 \times \{0\} \subset X^4 \times \mathbb{R}$, with $W = 0$; a local graph over it gives examples with $W \neq 0$.

Analytic Proof Route

Assume (g, W) is real analytic.

- A GKS forces the constraint equations required of a hypersurface in a Ricci-flat manifold:
$$d \operatorname{tr} W + \delta W = 0, \qquad \mathrm{Scal}_g + \operatorname{tr}(W^2) - (\operatorname{tr} W)^2 = 0.$$
- [Koi81]: these constraints allow us to use (g, W) as initial data for the Ricci-flat Einstein equation. Locally one constructs
$$Z = (-\varepsilon, \varepsilon) \times M, \qquad \bar{g} = dt^2 + g_t, \qquad \mathrm{Ric}_{\bar{g}} = 0,$$
so that $g_0 = g$ and the slice $\{0\} \times M$ has shape operator W.
- [AMM13]: Each $\psi \in \mathcal{K}_W(M)$ extends to a parallel spinor on the same Ricci-flat germ:
$$\mathcal{K}_W(M) \hookrightarrow \mathcal{P}(Z).$$

Thus k common GKS give k parallel spinors on one Ricci-flat ambient space, and

$$k > 2^{\lfloor n/2 \rfloor - 1} \overset{\text{Wang}}{\Longrightarrow} Z \text{ flat} \implies (g, W) \text{ Euclidean hypersurface data.}$$

Smooth Proof Route

- Curvature of ∇^W: for every $\psi \in \mathcal{K}_W(M)$,

$$\big(\widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y)\big) \cdot \psi = 0.$$

- Adjoin a formal unit normal e_0; the two defects become one bivector,

$$A_{X,Y} := \widehat{\mathcal{G}}_W(X,Y) + \widehat{\mathcal{C}}_W(X,Y) \wedge e_0 \ \in \Lambda^2\big(T_pM \oplus \mathbb{R}e_0\big).$$

- In orthogonal normal form $A = \sum_j \lambda_j\, e_{2j-1} \wedge e_{2j}$, the spinorial kernel is counted by the signed sums

$$\sum_j \varepsilon_j \lambda_j = 0, \qquad \varepsilon_j = \pm 1.$$

Two coefficient patterns govern the extremes:

$(\lambda, \lambda, 0, \ldots, 0)$: exactly half rank (the X-examples);
$(\lambda, \lambda, \ldots, \lambda)$: Kähler; eliminated by the Bianchi identity for $\mathcal{G}_W$

Dimensional Picture

Dimension	Threshold	Behaviour
2	1	Friedrich's surface correspondence
3	none	full fixed-sign spinor rank can still leave non-zero Gauss defect
$n \geq 4$	$2^{\lfloor n/2 \rfloor - 1} + 1$	sharp forcing threshold

The sharp half-rank example:

$$M^n = X^4 \times S^{n-4}(r) \subset X^4 \times \mathbb{R}^{n-3}, \qquad \mathrm{Hol}(X) = Sp(1).$$

One additional common generalised Killing spinor forces the Euclidean Gauss–Codazzi equations.

References

[AC09] D. V. Alekseevsky and V. Cortés, *On pseudo-Riemannian manifolds with many Killing spinors.*

[AMM13] B. Ammann, A. Moroianu and S. Moroianu, *The Cauchy problems for Einstein metrics and parallel spinors.*

[Art25] D. Artacho, *Generalised Killing spinors on three-dimensional Lie groups*, Manuscripta Math. 176 (2025).

[Bär93] C. Bär, *Real Killing spinors and holonomy.*

[Fri98] T. Friedrich, *On the spinor representation of surfaces in Euclidean 3-space.*

[Koi81] N. Koiso, *Hypersurfaces of Einstein manifolds.*

[LR10] M.-A. Lawn and J. Roth, *Isometric immersions of hypersurfaces in 4-dimensional manifolds via spinors.*

[Wan89] M. Y. Wang, *Parallel spinors and parallel forms.*